

WASTEWISE SCAN · CALCULATION

WasteWise Scan sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); decisions recorded in WSC-DDR-003 applied (sun hood, backing step and dark-level prompt, open-air crosstalk reading, modulation check)
0.1	2026-09-25	Amish Chadha	First issue for TRL 3 (mass, band coverage, signal and noise, sunlight, timing, power, logging, eye safety, display, drop and drift, cost)

WasteWise Scan sizing calculations

On paper, WasteWise Scan now meets nine of its fourteen requirements, has three at risk and none not met; two cannot be verified at TRL 3. This revision applies the decisions Amish accepted on 2026-09-25 (WSC-DDR-003). Sunlight (R4) moves from not met to met: clear and translucent items are backed with the black calibration cap, which cuts the ambient error to 0.36 % of signal, and a firmware rule refuses a result, rather than guessing, when the dark level shows light through an unbacked item. Display legibility (R7) moves from not met to at risk: a clip-on sun hood gives about 5.2:1 contrast where it shades the screen, but the screen still washes out with the sun near its normal or over the open side. Resin accuracy (R1) stays at risk because the 1,650 nm band reaches only the short edge of the 1,650 to 1,750 nm polymer region, and the shared record (R13) stays at risk until WasteWise-ml adopts it. The hood adds about 7 g and \$1: mass rises from 202 g to 209 g and the prototype cost from \$163.00 to \$164.00. The calculations at v0.1 changed two parts of the TRL 2 concept: the transimpedance gain dropped from 1 M Ω to 47 k Ω and each band reading sits between two dark readings. Every number in this note is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [C4], is the line of that script's output that carries it.

Safety: These are first-principles estimates for a paper proof of concept. They do not replace measurements of LED emission, drop behavior, cell temperature or classification accuracy. The scanner identifies resin only and must never be used to judge whether waste is safe to handle. See WSC-PRC-001, Safety.

■ Scope and method

The note checks every requirement in WSC-REQ-001 v0.3 against the design in WSC-PRC-001 v0.3 and the parametric model `cad/src/model.py`. The script imports the model's `PARAMS`, `derived()` and part solids, so the shells, shroud, window and optics geometry used here are those in the STEP files and in drawing WSC-DWG-001. It also reads `bom/bom.csv` and `budget_usd` in `project.yaml`. Run it from the repo root with `python docs/04-calcs/sizing.py`.

The design case is the decided configuration (WSC-DDR-001 and WSC-DDR-003): eight LEDs from 850 to 1,650 nm, one 1 mm InGaAs photodiode, an ADS1220-class 24-bit ADC, a T-Display-S3-class board, one protected 3,000 mAh 18650 cell and an 18 mm shroud that puts the item 22 mm from the detector [C1], a black calibration cap that doubles as a backing for clear items, and a clip-on sun hood 20 mm high over the display.

■ Assumptions

Table 1. Main assumptions. All are to be confirmed from datasheets or measurement.

Area	Assumption	Basis
LEDs	Radiant power at 100 mA: 15 mW at 850 and 940 nm, 4 mW at 1,050 nm, 2 mW at 1,200 and 1,300 nm, 1.5 mW at 1,450 and 1,550 nm, 1 mW at 1,650 nm. FWHM 40 to 120 nm, widening with wavelength. Beam half-angle 10 to 15°. Output falls 0.4 %/°C	Typical catalog ranges for small IR and SWIR LEDs; not from a chosen part
Detector	Standard InGaAs, 1 mm active diameter; responsivity 0.35 A/W at 900 nm, 0.90 at 1,300 nm, 0.95 at 1,550 nm, 0.50 at 1,680 nm, 0.05 at 1,720 nm, linear between	Typical curve shape; confirm from the chosen part
Optics	Uncoated borosilicate window, 0.92 transmission per pass, 4 % reflection per surface; illuminated spot 8 mm radius; item a Lambertian reflector	Standard values for uncoated glass; Lambertian is a first-order model
Items	Diffuse reflectance 0.9 for the PTFE reference, 0.5 for a typical opaque item, 0.10 for a clear item, 0.04 for a carbon-black item. Diffuse transmittance of a clear or translucent wall 0.30 (worst case)	Assumed; to measure on reference chips
Electronics	ADC at 330 samples/s, 5 μ V rms per conversion, 2.048 V range; 4 conversions averaged; 2 ms LED settling; 3.3 V amplifier rail	ADS1220-class figures, conservative
Sunlight	300 W/m ² in the 900 to 1,700 nm band at about 100 klx; hand movement changes ambient light by 1 % per 10 ms	Share of the solar spectrum, rounded; movement figure assumed
Power	ESP32-S3 at 240 MHz with radio off 45 mA, display at full backlight 60 mA, analog front end 3 mA; LEDs 100 mA while on; 4 scans per minute; 80 % of 10.8 Wh usable	Typical values; to measure on the board
Display	400 cd/m ² white, 0.4 cd/m ² black, 2 % diffuse-equivalent screen reflectance; about 3:1 contrast needed for large colored icons	Not from a datasheet; legibility threshold assumed
Drop	1.2 m; 1 mm crush on a shell corner, 10 mm on the TPU shroud	First-order stopping distances
Decided options	Black cap leaves 10 ⁻³ of the through-light when backing a clear item; ambient error limit 0.5 % of signal; window crosstalk changes by 10 % between open-air readings; modulation check at 2 kHz	Assumed; cap opacity and crosstalk stability to measure at TRL 4

■ A. Geometry and mass (R6)

The scanner weighs about 209 g, and 229 g with the calibration cap [A3], against 250 g. The printed shells are the largest share (48 g and 47 g from the model volumes in PETG), followed by the cell (47 g), the display board (15 g) and the sun hood (7 g) [A2]. The TRL 2 estimate of about 170 g undercounted the shells. The body is 160 x 62 x 34 mm (6.3 x 2.4 x 1.3 in); the 18 mm shroud below (rim 42 mm outside and 36 mm inside) and the 20 mm hood above make it 72 mm high overall [A1]. R6 is met; its text now names the hood alongside the shroud as an attachment outside the body envelope (WSC-REQ-001 v0.4).

■ B. Band coverage and detector response (R1)

Seven of the eight bands sit where a standard InGaAs detector responds well, but the 1,650 nm band is clipped by the detector's cutoff: its detected centroid moves down to about 1,628 nm and only about 31 % of its detected signal lies at 1,650 nm or above [B1, B2]. Responsivity falls to 0.28 A/W at 1,700 nm and 0.05 A/W at 1,720 nm [B2]. Only the 1,200 nm band falls inside the 1,150 to 1,250 nm C-H second-overtone region; the 1,650 to 1,750 nm first-overtone region is reached only at its short edge [B3]. The 850 nm band has a mean responsivity of only 0.20 A/W, which the brighter 850 nm LED offsets [B1].

This supports the TRL 2 concern that PE versus PP, which relies on features near 1,700 nm, is the weak point. R1 is **at risk**. It cannot be verified without reference spectra and a labeled item set, and the band study that would allow six bands (WSC-DDR-001, D2) could not be run without reference spectra, so the design keeps eight bands. Whether to add a 1,700 to 1,750 nm band on an extended InGaAs detector remains open (WSC-DDR-001, O3).

■ C. Optical signal, crosstalk and noise (R1, R3)

In shade, every band has a signal-to-noise ratio of about 450 or more even on a clear, weakly reflecting item, against about 200 needed to resolve 0.5 % reflectance differences [C4]. The Lambertian collection factor is 4.6×10^{-4} per unit reflectance [C1]. On the white reference, the detected power ranges from 0.35 μ W at 1,650 nm to 5.2 μ W at 850 and 940 nm, giving 10 to 115 mV at 47 k Ω [C2]. The TRL 2 figure of 0.1 to 1 μ A stands; the strongest band gives 2.45 μ A [C3].

Two findings matter more than noise:

- **Window crosstalk.** LED light reflected by the window's item-side surface can reach the detector through the glass under the baffle tube. The path is 13.8 mm, and the crosstalk is of the same order as the signal from a 0.5-reflectance item [C5]. It is a fixed offset per band that an open-air reading can remove, but dirt or scratches on the window change it. Amish accepted option (a) on 2026-09-25 (WSC-DDR-003 D11): the calibration routine takes an open-air reading and subtracts the offset. Because the offset is as large as the signal, a 10 % change from window dirt leaves about 10 % of the signal as error; to stay under 0.5 % the offset must stay within 0.5 % of itself between open-air readings [C7]. That is unlikely on a dirty window, so the decided fallback, a baffle extended through the glass with separate LED and detector windows (option (b)), is probably needed. Whether it is needed can only be shown by measurement at TRL 4.
- **Black items.** A carbon-black item returns about 8 % of a typical item's signal. A threshold at mean reflectance 0.08 separates it with a signal-to-noise ratio of about 180, so black items read "unknown" reliably [C6]. The wrong-result rate in R3 needs labeled item data, so R3 is **not verifiable at TRL 3**.

■ D. Sunlight and stray light (R4)

The shroud stops sunlight falling on the spot, but not light passing through the item from behind. For a clear or translucent item in direct sun, light through the item fills the 36 mm rim with a view factor of 0.40 and puts about 28 μ W, or 22.7 μ A, on the detector [D1]. The TRL 2 gain of 1 M Ω would saturate at 22.7 V; 47 k Ω keeps the output at 1.07 V, inside the 2.048 V ADC range [D2], and the design now uses 47 k Ω . The shot-noise-limited signal-to-noise ratio is still about 430 [D4].

The limit is change in the ambient light, not noise. Ambient is about 936 times the 1,650 nm signal from a clear item, so a 1 % change per 10 ms from hand movement leaves about 7.5 mV after interleaved dark subtraction, against 1.14 mV of signal [D3]. On an opaque item the rim leak is small and the same error is 0.13 % of signal [D5]. Without a change, R4 would not be met for clear and translucent items in direct sun. Amish accepted option (b) on 2026-09-25 (WSC-DDR-003 D8 and D9):

- **Backing step.** The user backs a clear or translucent item with the calibration cap, which is now printed in black PETG. The PTFE disc sits behind the wall, so the scanner reads light that passes through the wall and back (effective reflectance about 0.18), and the black cap blocks sunlight from behind. With 10^{-3} of the through-light left, the drift error is 7.5 μ V against 2.06 mV, or 0.36 % of signal, inside the 0.5 % limit [D6]. The classifier needs training data taken in this backed mode.
- **Dark-level prompt.** The firmware shows "shade or back with cap" instead of a result when the mean dark level exceeds 0.71 times the weakest band reading, the point at which the drift error reaches 0.5 %. An unbacked clear item in sun sits at about 936 times and is refused; a backed item (0.52) and an opaque item (0.19) get a result [D7].
- **Modulation check (option (a), paper only).** Pulsing the LEDs at 2 kHz with synchronous demodulation shortens the light-to-dark gap to 0.25 ms, but an unbacked clear item in sun would still carry an error of about 23 % of the 1,650 nm signal, and the ADC would need 8 kS/s or more or an analog demodulator [D8]. Option (a) therefore does not replace the backing step; restating R4 (option (c)) is not needed.

With these, R4 as restated in WSC-REQ-001 v0.4 is **met by estimate**: opaque items carry 0.13 % error, backed clear items 0.36 %, and unbacked clear items in sun are refused rather than misread.

■ E. Scan timing (R2)

A reading takes 14.1 ms: 2 ms of settling plus four conversions of 3.03 ms. Eight bands and nine interleaved dark readings take 240 ms [E1]. With classification, display update and debounce, button to result is about 0.33 s against 1.5 s [E2]. R2 is met.

■ F. Power and battery (R5)

The average draw is about 109 mA (0.39 W), dominated by the display backlight and the microcontroller; the LEDs use about 42 mJ per scan, or 0.77 mA on average [F1, F2]. The 10.8 Wh cell, 80 % usable, lasts about 22.1 h against 8 h [F2]. A USB charge takes about 7.2 h at an assumed 500 mA [F3]. R5 is met.

■ G. Log storage (R9)

At 64 bytes per record (layout in WSC-DDR-002), 10,000 scans take 625 kB, and an 8 MB log partition of the 16 MB flash holds about 131,000 scans. A CSV export of 10,000 rows is about 1.4 MB [G1]. R9 is met.

■ H. Eye safety (R14)

The emission stays well inside the IEC 62471 exempt group by estimate. The brightest LED (850 nm, 15 mW, $\pm 10^\circ$) has a radiant intensity of about 157 mW/sr [H1]. At the 200 mm reference distance, corneal irradiance is 3.9 W/m², against an exempt limit of 100 W/m²; at the shroud rim it would be about 241 W/m² if an LED were held on continuously, and about 0.23 W/m² averaged over normal use [H2]. The weighted retinal thermal radiance is about 3.8 % of the exempt limit [H3]. R14 is met by estimate. Because a stuck LED at the rim would exceed the corneal

limit over long exposures, the LED drivers need a hardware limit on on-time (see WSC-PRC-001, Safety). The limit values are those of IEC 62471:2006 as understood here; confirm them against the standard and the LED datasheets.

■ I. Display legibility (R7)

In 100 klx sun, the screen reflects about 637 cd/m² over a 400 cd/m² image, giving about 1.6:1 contrast against about 3:1 needed [I1]. Shaded from direct sun (about 15 klx of sky light), contrast is about 5.2:1 [I2]. Amish accepted option (a) on 2026-09-25 (WSC-DDR-003 D10): a clip-on sun hood with walls 20 mm high on both sides and at the head end of the 46 x 24 mm window, open toward the user. The screen is fully shaded when the sun is at least 50° from the screen normal on a side, or 67° on the head end. At 45° the hood shades 83 % of the screen from the side and 43 % from the head end; at 30°, 48 % and 25 % [I3]. The shaded part reads at about 5.2:1; with the sun near the screen normal or over the open side, contrast stays about 1.6:1 [I4]. Large color and icon cues remain the main result. R7 moves from not met to **at risk**: legibility in direct sun now depends on how the user holds the scanner relative to the sun.

■ J. Drop and temperature (R10, R11)

A 1.2 m drop delivers about 2.4 J. On a shell corner with about 1 mm of crush, deceleration is about 1,200 g; landing on the TPU shroud, about 120 g [J1]. The cell cradle must then hold the cell against about 550 N and the display board against about 180 N [J2]. IP54 and the 0 to 45 °C range depend on the gasket and the cell, so R10 is **not verifiable at TRL 3**.

LED output drifts about 0.4 %/°C, so the 5 % drift warning in R11 trips after about 12.5 °C of change since the last white reference [J3]. A morning calibration at 25 °C will need repeating by a hot midday. The cap makes this a 10 s step, so R11 is met by design.

■ K. Cost (R12)

The priced BOM has 16 lines and totals \$164.00, 9.3 % over the \$150 volume target in `budget_usd`; the LEDs and photodiode are \$92.00, or 56 % [K1]. Amish accepted about \$163 for the first prototype on 2026-09-25 (WSC-DDR-001, D2) and then the sun hood (WSC-DDR-003 D10), which adds \$1.00 of PETG, so R12 as restated in WSC-REQ-001 v0.4 (about \$164 for the prototype) is met. The \$150 volume target cannot be shown without volume quotes.

■ Results

Table 2. Requirement status at TRL 3, weakest first [L1, L2]. None is not met.

ID	Requirement	Value	Target	Status
R1	Identify common resins	8 bands; 1,650 nm band centroid 1,628 nm, 31 % of its signal at or above 1,650 nm	95 % correct on 5 resins	At risk
R7	Readable by anyone	5.2:1 where the hood shades the screen; 1.6:1 with the sun near the screen normal or over the open side	Legible in direct sun (about 3:1)	At risk (was not met)
R13	Work with WasteWise-ml	Record decided on this side (WSC-DDR-002 v0.2); not yet adopted by WasteWise-ml	Shared documented format	At risk
R3	Fail safe on unreadable items	Black items fall below the 0.08 threshold; wrong-result rate needs item data	2 % or fewer wrong	Not verifiable at TRL 3
R10	Rugged	Corner drop about 1,200 g; cell needs about 550 N retention	1.2 m drop, IP54, 0 to 45 °C	Not verifiable at TRL 3
R2	Fast result	0.33 s	1.5 s or less	Met
R4	Work in sunlight	Opaque 0.13 %, backed clear 0.36 % of signal; unbacked clear items in sun refused by the dark-level prompt	Same result in 100 klx sun; clear items backed with the cap	Met (estimate; was not met)
R5	Last a shift	22.1 h	8 h or more	Met
R6	One-handed and light	209 g (229 g with cap); 160 x 62 x 34 mm	250 g; 170 x 65 x 40 mm plus shroud and hood	Met
R8	Local price grade	Price table on the device, offline	Editable local grades	Met (by design)
R9	Log scans	About 131,000 scans	10,000 scans	Met
R11	Stay calibrated	Cap reference and open-air reading; 5 % drift after about 12.5 °C	Check in 10 s; warn at 5 %	Met (by design)
R12	Low cost	\$164.00 prototype	About \$164 prototype (accepted); \$150 volume target	Met (prototype)
R14	Eye-safe illumination	3.9 W/m ² at 200 mm; 3.8 % of the retinal limit	IEC 62471 exempt	Met (estimate)

Summary: none not met, 3 at risk (R1, R7, R13), 2 not verifiable at TRL 3 (R3, R10), 9 met.

■ Checks against earlier documents

- Scan time: the TRL 2 figure of about 0.6 s assumed 15 ms per conversion; with 330 samples/s and interleaved darks it is about 0.33 s. WSC-PRC-001 v0.3 now uses 0.33 s.
- Battery life: about 17 h at TRL 2 (0.5 W); about 22 h here (0.39 W). The precis now uses about 22 h.
- Mass: about 170 g at TRL 2; 202 g here from model volumes. The precis now uses 202 g.
- Photocurrent: the TRL 2 range of 0.1 to 1 µA is confirmed for most bands; the brightest reaches about 2.5 µA.
- Cost: \$163.00 at v0.1; \$164.00 at v0.2 with the sun hood.
- Mass: 202 g at v0.1; 209 g at v0.2 with the sun hood.